

DDoS Threat Analysis — Five Recent Incidents

Objective

Analyze the threat of five recent DDoS attacks. For each incident provide: target, technology used, attacker's motive (inference), overall impact, and defensive strategies. A focused investigation is included for one selected incident.

Incident 1 — Cloudflare mitigated 7.3 Tbps DDoS (May 2025)

Target: A Cloudflare customer (a hosting provider using Cloudflare Magic Transit).

Technology / Vector: Multivector, predominantly UDP floods at massive scale (targeting tens of thousands of destination ports), small reflection/amplification components (e.g., QOTD, Echo, NTP). Originating from >100k source IPs across thousands of ASNs.

Attacker's Motive (inference): Unknown public attribution; likely opportunistic capacity-test, capability demonstration, or ransom/extortion motive in some similar campaigns.

Overall Impact: Peak ~7.3 Tbps, ~37.4 TB delivered in ~45 seconds; mitigated by Cloudflare's anycast + scrubbing and automated eBPF/XDP filtering. Unprotected networks could have faced severe outages.

Defensive Strategies / Mitigations: Cloud-based scrubbing/anycast, kernel-level sampling (XDP/eBPF), BCP38 egress filtering to limit spoofing, hardening UDP reflection services, rate-limiting and scrubbing centres.

Incident 2 — Aisuru botnet attacks on ISPs / gaming infrastructure (Oct 2025)

Target: Gaming platforms, TCPShield (Minecraft DDoS protection), and various ISPs/hosting providers.

Technology / Vector: Large IoT-based botnet volumetric UDP/TCP floods (reports of 15 Tbps+ peak during focused blasts), carpet-bombing many ports and saturating ISP links.

Attacker's Motive (inference): Likely motivated by disruption (competition/crippling services) and possibly ransom/extortion or capability demonstration.

Overall Impact: Widespread connectivity failures for gaming communities and overloaded ISP peering links; mitigation required coordination with upstreams and scrubbing.

Defensive Strategies / Mitigations: DDoS scrubbing services, traffic engineering with anycast, cooperation with transit providers, filtering at edge, botnet takedowns and IoT hygiene.

Incident 3 — HTTP/2 Rapid Reset and high-RPS campaigns (Jan 2025 and early 2025)

Target: Retail, government, and corporate web applications (examples: French retail site, Indonesian government agency, others).

Technology / Vector: Application-layer attacks using HTTP/2 Rapid Reset technique (forcing servers to allocate resources and reset streams), combined with high-RPS floods across distributed geolocations.

Attacker's Motive (inference): Often politically or financially motivated disruption, or automated extortion; sometimes used to test amplification of HTTP/2 weaknesses.

Overall Impact: High request rates (millions of RPS in observed cases), hours-long outages or degraded service, difficulty filtering because traffic mimics legitimate HTTP/2 patterns.

Defensive Strategies / Mitigations: WAF protections tuned for HTTP/2 anomalies, rate-limiting per-client, connection limits, CAPTCHA/JS challenges for suspicious clients, and anomaly-based behavioral detection.

Incident 4 — Near-record 6.3 Tbps DDoS against KrebsOnSecurity (May 2025)

Target: KrebsOnSecurity (independent security news site).

Technology / Vector: Massive IoT botnet UDP floods (mirroring earlier Mirai-like botnet iterations) producing terabit-scale volumetric traffic.

Attacker's Motive (inference): Possibly a capability test by a new IoT botnet operator or an extortion attempt; suspected IoT compromised devices used as a platform.

Overall Impact: Brief but extremely large volumetric spike (6.3 Tbps) causing temporary disruption potential for non-mitigated networks; served as a wake-up call to many orgs.

Defensive Strategies / Mitigations: ISP-level filtering, upstream scrubbing, IoT device hygiene and takedowns, and collaboration with hosting providers for scrubbing.

Incident 5 — GitHub global outage (July 28-29, 2025)

Target: GitHub core services (Issues, API, Pull Requests) experienced global disruption.

Technology / Vector: Public reports suggested large-scale traffic disruption; some community writeups discussed DDoS-like symptoms though GitHub's official cause summary focused on network/service disruption.

Attacker's Motive (inference): If DDoS-linked, motive may be extortion or opportunistic disruption; if operational, then no attacker motive.

Overall Impact: Service disruption for developers and CI/CD pipelines for several hours; economic and productivity impact for many organizations.

Defensive Strategies / Mitigations: For platform providers: multi-region redundancy, autoscaling, robust traffic isolation for control-plane APIs, and rapid incident communication.

Incident 6 — Focused Investigation — Cloudflare 7.3 Tbps DDoS (May 2025)

Selected because of record scale, transparent vendor writeup, and clear mitigation lessons.

Target: A customer of Cloudflare (hosting provider) whose IPs were protected by Cloudflare Magic Transit.

Technology / How the attack worked:

Multivector; predominantly UDP floods aimed at tens of thousands of destination ports. Small amounts of reflection/amplification (QOTD, Echo, NTP) were observed. The attack was highly distributed (>100k source IPs, thousands of ASNs).

Attacker motive (inference): Public reports did not attribute the attack to a named actor. Probable motives include capability demonstration, testing a botnet, or opportunistic/extortion activity.

Overall impact:

Peak 7.3 Tbps and ≈ 37.4 TB delivered in ~ 45 seconds. The attack was mitigated by Cloudflare's global anycast and automated filtering; had the target been unprotected it could have saturated upstream links and caused prolonged outages.

Defensive strategies that mitigated the attack:

- Cloud-based scrubbing and anycast distribution: Dissipate volumetric traffic across many points of presence.
- Kernel-level sampling and automated XDP/eBPF fingerprinting: Create precise drop rules quickly.
- Blocking/rate-limiting UDP reflection vectors and BCP38 egress filtering: Reduce spoofed-source amplification.
- Pre-established runbooks and transit provider coordination: Enable rapid mitigation.
- Continuous monitoring, threat intelligence feeds, and automated mitigation pipelines: Ensure rapid response to threats.